

SILICON UNIVERSITY

NAME- ARADHANA MOHANTY

SIC-24MMCD89

DATE OF SUBMISSION -20/07/2025

Acknowledgement:

I am grateful to have successfully completed my summer internship at **Ingenious-tech** as part of my Master's program, where I gained valuable hands-on experience in Microsoft Azure. This internship provided me with deep insights into cloud computing, virtual networking, and full-stack deployment. I am now pleased to submit my final project, which is based on the practical knowledge and skills I acquired during this enriching learning journey.

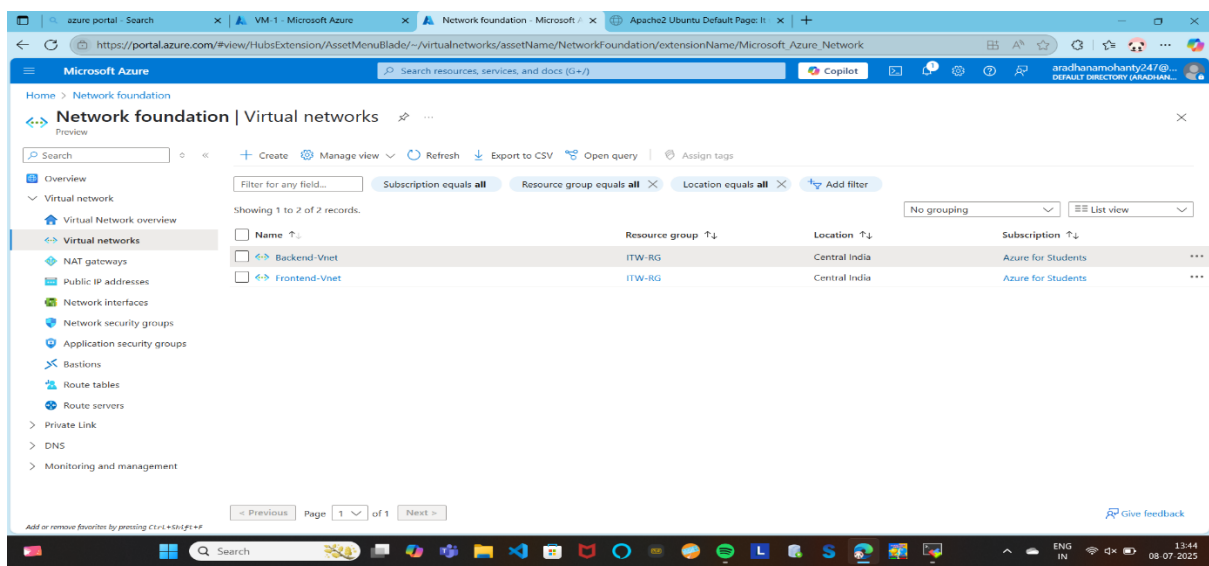
AZURE CLOUD ASSIGNMENT

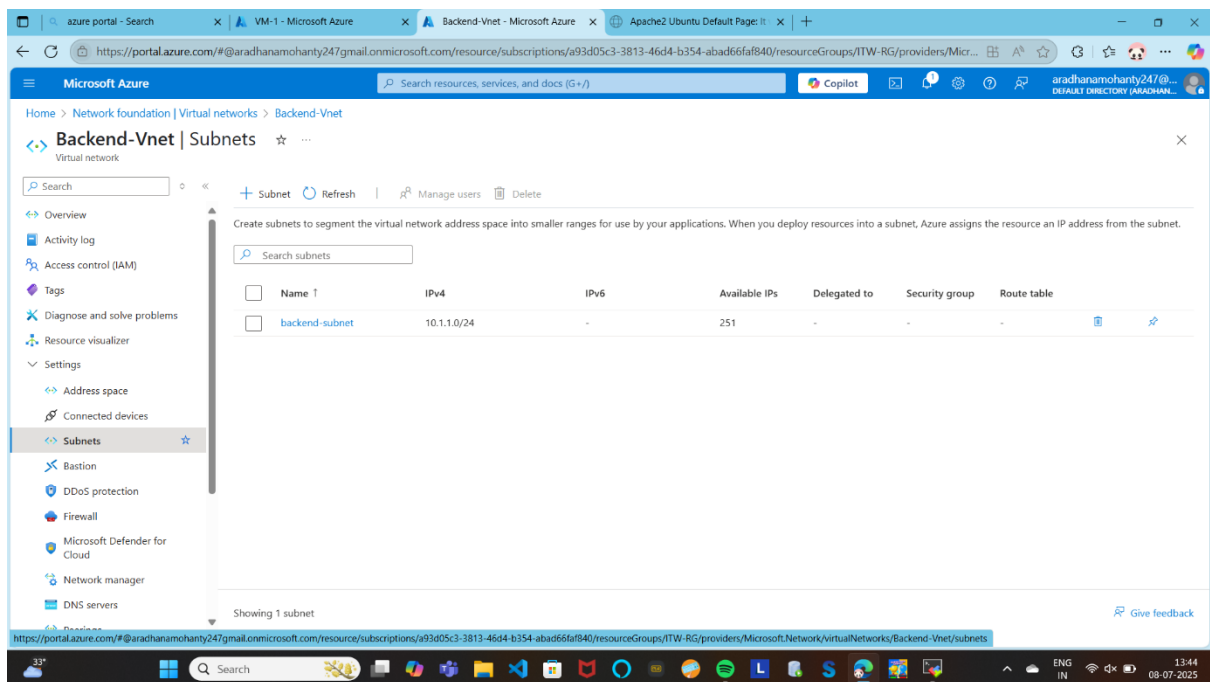
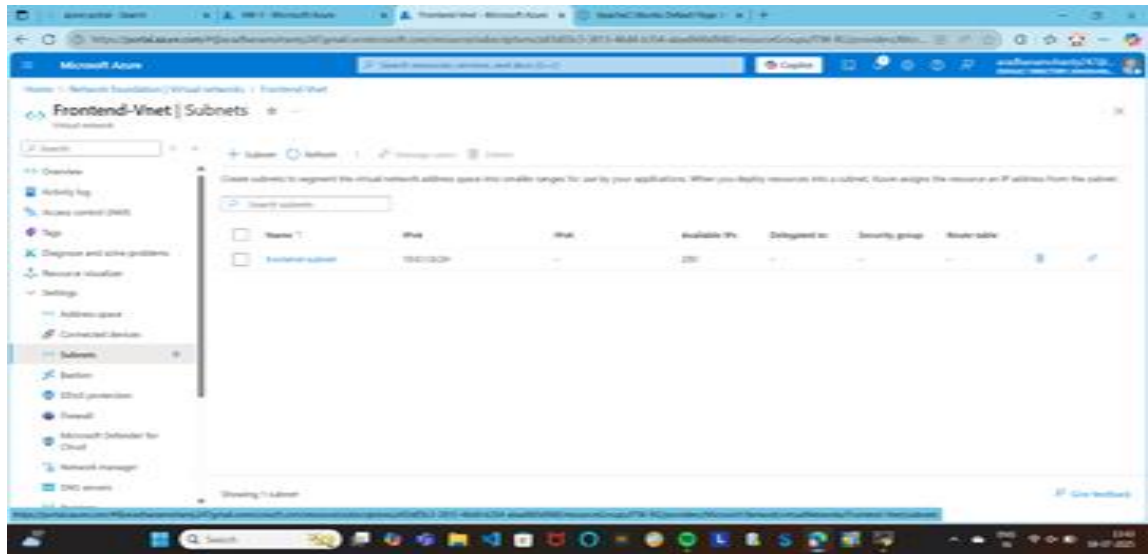
Full -Stack App Deployment with VNet, VM, Load Balancer Domain

Step 1- Create separate Virtual Network (VNet) for the frontend and backend /database parts of the application:

Definations:

- **Virtual Network (VNet):** A Virtual Network in Azure is a private, isolated network that allows Azure resources like VMs to securely communicate with each other, the internet, and on-premises networks.
- **Subnet:** A subnet is a segment within a virtual network where specific resources like virtual machines are placed. Subnets help organize and isolate resources for better management and security.
- **Public IP:** A public IP address is used to allow external access to a virtual machine over the internet. It makes the VM reachable from outside the Azure network.





Step 2- Deploying Virtual Machines inside each VNet:

- *One (or more) for the frontend*
- *One for the database/backend*
- *Include a Jump VM to security access both networks and perform changes.*

In Step 2, we deployed virtual machines (VMs) within separate virtual networks (VNets) for frontend and backend operations. A Fronted-VM was created inside the frontend VNet to handle web server tasks. Another Backend- VM was deployed in the backend VNet for hosting the database. A Jump VM was also set up in a dedicated subnet with a public IP to securely access both frontend and backend VMs. To enable secure communication between VNets, we configured VNet peering between the frontend and backend VNets. This setup ensures proper isolation, security, and connectivity across different layers of the application.

Definations:

- Virtual Machine (VM): A virtual machine is a software-based emulation of a physical computer. It allows you to run applications, host services, and perform computing tasks without the need for physical hardware.
- Private IP: A private IP address is only accessible within the Azure virtual network. It's used for secure, internal communication between resources.
- Jump Server (or Bastion Host): A jump server is a virtual machine with a public IP that acts as a secure entry point to access other VMs in private subnets or VNets that do not have public access.
- VNet Peering: VNet peering connects two virtual networks in Azure, allowing resources in those VNets to communicate with each other using private IP addresses over Azure's backbone network.
- SSH (Secure Shell): SSH is a secure protocol used to remotely access and manage Linux-based systems through a command-line interface.
- NSG (Network Security Group): NSGs are firewall-like rule sets that control the flow of network traffic in and out of Azure resources such as VMs or subnets.

azure portal - SearchFrontend-Vnet - Microsoft AzureCompute infrastructure - MicroApache2 Ubuntu Default Page: R x +

https://portal.azure.com/#view/Microsoft_Azure_ComputeHub/ComputeHubMenuBlade/~virtualMachinesBrowse

Microsoft AzureSearch resources, services, and docs (G+)

Copilot

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Home > Compute infrastructure

Compute infrastructure | Virtual machines

Virtual machinesGet started

OverviewAll resourcesInfrastructureVirtual machinesVirtual Machine Scale Set (VMSS)Compute FleetDisks + imagesCustom imagesDisksSnapshotsDisk encryption setsCapacity + placementRelated servicesHelp

+ CreateSwitch to classicReservationsManage viewRefreshExport to CSVOpen queryAssign tagsStartGroup by none

You are viewing a new version of Browse experience. Some features may be missing. Click here to access the old experience.

Filter for any field...Subscription equals allType equals allResource Group equals allLocation equals allAdd filter

	Name ↑	Subscription	Resource Group	Location	Status	Operating syst...	Size	Public IP addre...	Disks
☐	Backend-VM	Azure for Stude...	ITW-RG	Central India	Running	Linux	Standard_D2s_v3	-	1
☐	Jump-VM	Azure for Stude...	ITW-RG	Central India	Running	Linux	Standard_B1s	98.70.42.251	1
☐	VM-1	Azure for Stude...	ITW-RG	Central India	Running	Linux	Standard_D2s_v3	4.240.93.5	1

Showing 1 - 3 of 3. Display count: 10

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https://portal.azure.com/#@aradhanamohanty247gmail.onmicrosoft.com/resource/subscriptions/a93d05c3-3813-46d4-b354-abad66faf840/resourceGroups/ITW-RG/providers/Mi...

Microsoft AzureSearch resources, services, and docs (G+)

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Home > Network foundation | Virtual networks > Frontend-Vnet

Frontend-Vnet | Subnets

Virtual network

Search

+ SubnetRefreshManage usersDelete

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

	Name ↑	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table
☐	frontend-subnet	10.0.1.0/24	-	250	-	-	
☐	jump-server	10.0.2.0/24	-	250	-	-	

Showing 2 subnets

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https://portal.azure.com/#@aradhanamohanty247gmail.onmicrosoft.com/resource/subscriptions/a93d05c3-3813-46d4-b354-abad66faf840/resourceGroups/ITW-RG/providers/Microsoft.Network/virtualNetworks/Frontend-Vnet/subnets

15:4308-07-2025

azure portal - Search Frontend-Vnet - Microsoft Azure Frontend-Vnet - Microsoft Azure Apache2 Ubuntu Default Page: It x +

https://portal.azure.com/#@aradhanamohanty247gmail.onmicrosoft.com/resource/subscriptions/a93d05c3-3813-46d4-b354-abad66faf840/resourceGroups/ITW-RG/providers/Mi...

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Home > Network foundation > Virtual networks > Frontend-Vnet

Frontend-Vnet | Peerings

Virtual network

Search

+ Add Refresh Export to CSV Delete Sync

Virtual network peering enables you to seamlessly connect two or more virtual networks in Azure. The virtual networks appear as one for connectivity purposes. [Learn more](#)

Filter by name...

Showing all 1 items

☐	Name	Peering sync status	Peering state	Remo...	Virtu...	Cross-tenant
☐	BackendToFrontend	Fully Synchronized	Connected	Backend...	Disabled	No

Give feedback

https://portal.azure.com/#@aradhanamohanty247gmail.onmicrosoft.com/resource/subscriptions/a93d05c3-3813-46d4-b354-abad66faf840/resourceGroups/ITW-RG/providers/Microsoft.Network/virtualNetworks/Frontend-Vnet/peerings

azure portal - Search Frontend-Vnet - Microsoft Azure Backend-Vnet - Microsoft Azure Apache2 Ubuntu Default Page: It x +

https://portal.azure.com/#@aradhanamohanty247gmail.onmicrosoft.com/resource/subscriptions/a93d05c3-3813-46d4-b354-abad66faf840/resourceGroups/ITW-RG/providers/Mi...

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Home > Network foundation > Virtual networks > Backend-Vnet

Backend-Vnet | Peerings

Virtual network

Search

+ Add Refresh Export to CSV Delete Sync

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Filter by name...

Showing all 1 items

☐	Name	Peering sync status	Peering state	Remo...	Virtu...	Cross-tenant
☐	FrontendToBackend	Fully Synchronized	Connected	Fronten...	Disabled	No

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https://portal.azure.com/#@aradhanamohanty247gmail.onmicrosoft.com/resource/subscrip...

S

```
siliconuser123@Backend-VM: ~  
Terminal Sessions View X server Tools Games Settings Macros Help  
Session Servers Tools Games Sessions View Split Multitask Tunneling Packages Settings Help  
Quick co 2 siliconuser123@Backend-VM -  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Failed to add the host to the list of known hosts (/home/mca/.ssh/known_hosts).  
siliconuser123@98.70.42.251's password:  
Permission denied, please try again.  
siliconuser123@98.70.42.251's password:  
client_input hostkeys: hostkeys.foreach failed for /home/mca/.ssh/known_hosts: Permission denied  
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.11.0-1017-azure x86_64)  
  
 * Documentation:  https://help.ubuntu.com  
 * Management:    https://landscape.canonical.com  
 * Support:        https://ubuntu.com/pro  
  
System information as of Tue Jul 8 10:08:01 UTC 2025  
  
System load:  0.04      Processes:      112  
Usage of /:   5.5% of 28.02GB  Users logged in:  0  
Memory usage: 30%      IPv4 address for eth0: 10.0.2.4  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
siliconuser123@Jump-VM:~$ ssh siliconuser123@10.1.1.4  
UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net
```

```
siliconuser123@Backend-VM: ~  
Terminal Sessions View X server Tools Games Settings Macros Help  
Session Servers Tools Games Sessions View Split Multitask Tunneling Packages Settings Help  
Quick co 2 siliconuser123@Backend-VM -  
The authenticity of host '10.1.1.4 (10.1.1.4)' can't be established.  
ED25519 key fingerprint is SHA256:r588P6x4L6gJepPyJ5E+uM7MZ83IZR3JefsU8Lx20.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '10.1.1.4' (ED25519) to the list of known hosts.  
siliconuser123@10.1.1.4's password:  
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.11.0-1017-azure x86_64)  
  
 * Documentation:  https://help.ubuntu.com  
 * Management:    https://landscape.canonical.com  
 * Support:        https://ubuntu.com/pro  
  
System information as of Tue Jul 8 10:09:31 UTC 2025  
  
System load:  0.00      Processes:      133  
Usage of /:   5.5% of 28.02GB  Users logged in:  0  
Memory usage: 4%        IPv4 address for eth0: 10.1.1.4  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
siliconuser123@Backend-VM:~$  
UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net
```

Step 3-Attach a Load Balancer in front of the frontend VMs and assign a public IP to load balancer only (not the individual VMs).

In this project, an Azure Standard Public Load Balancer named frontend-lb was configured to distribute incoming traffic across multiple VMs in a backend pool using port forwarding and health probes. A frontend IP was set up to listen on port 77, which forwards traffic to port 80 on the backend VMs. A health probe on port 80 ensures only healthy VMs receive traffic. An outbound rule with SNAT was also configured to allow backend VMs internet access without needing public IPs. This setup ensures high availability, scalability, and efficient traffic distribution for the stack application.

Definitions:

- Azure Load Balancer: It is a Layer-4 (TCP, UDP) load balancer that distributes incoming traffic among healthy virtual machines (VMs) in a backend pool. It ensures high availability, scalability, and fault tolerance for your application.
- Frontend IP Configuration: The public IP that receives incoming traffic. In your case, traffic comes in on port 77
- Backend Pool: Contains the NICs of VMs (e.g., frontend, backend VMs). These VMs serve traffic.
- Health Probe: Monitors VMs on port 80. If a VM doesn't respond, it's removed from traffic routing.
- Load Balancing Rule: Routes external requests from port 77 (frontend) to port 80 (backend) of the VMs in the backend pool.
- Outbound Rule: Configures SNAT (Source Network Address Translation), so VMs without a public IP can still access the internet.

Microsoft Azure portal interface showing the "Frontend IP configuration" for a load balancer named "frontend-lb". The page includes a search bar, navigation pane, and a table listing the IP configuration.

Search: Add Refresh

The frontend IP address configuration of a load balancer serves as the entry point for incoming traffic to the load balancer, and the load balancer then distributes the traffic to the backend pool of virtual machines or services. [Learn more](#)

Type to start filtering ...

Showing all 1 items

Name	IP address	Rules count
frontend-ip	98.70.243.206 (frontend-lb-pip)	2

Give feedback

Microsoft Azure portal interface showing the "Backend pools" for a load balancer named "frontend-lb". The page includes a search bar, navigation pane, and a table listing the backend pools.

Search: Add Filter

The backend pool is a critical component of the load balancer. The backend pool defines the group of resources that will serve traffic for a given load-balancing rule. [Learn more](#)

Backend pool	Resource Name	IP address	Network interface	Availability zone	Rules count	Resource Status	Admin state
frontend-backend-pool (1)							
frontend-backend-pool	VM-1	10.0.1.4	vm-1656_z2	2	1	Running	None
frontend-vm-nic (1)							
frontend-vm-nic	-	-	-	-	1	-	-

Give feedback

Microsoft Azure portal interface showing the "Health probes" for a load balancer named "frontend-lb". The page includes a search bar, navigation pane, and a table listing the health probes.

Search: Add Refresh Give feedback

Type to start filtering ...

Name	Protocol	Port	Path	Used By
http-probe	Http	80	/ or /index.php	http-rule

Microsoft Azure

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Home > Microsoft.LoadBalancer-20250710123413 | Overview > frontend-lb

frontend-lb | Load balancing rules

Load balancer

Search

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Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Frontend IP configuration

Backend pools

Health probes

Load balancing rules

Inbound NAT rules

Outbound rules

Properties

Locks

Monitoring

Automation

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A load balancer rule is used to define how incoming traffic is distributed to the all the instances within the backend pool. A load-balancing rule maps a given frontend IP configuration and port to multiple backend IP addresses and ports. An example would be a rule created on port 80 to load balance web traffic. [Learn more.](#)

Filter by name...

Name	Protocol	Backend pool	Health probe	Health status
http-rule	TCP/77 to TCP/80	frontend-backend-pool	hhttp-probe	View details

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frontend-lb | Outbound rules

Load balancer

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Overview

Activity log

Access control (IAM)

Tags

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Settings

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Load balancing rules

Inbound NAT rules

Outbound rules

Properties

Locks

Monitoring

Automation

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Use outbound rules to configure the outbound network address translation (NAT) for all virtual machines in the backend pool. To create an outbound rule, the load balancer SKU must be standard and the frontend IP configuration must have at least one public IP address. [Learn more about outbound connectivity.](#)

Filter by name... Frontend IP address : all Backend pool : all Protocol : all

Name	Frontend IP address	Backend pool	Protocol	Ports per insta...	Delete
outbound-rule	frontend-ip	frontend-vm-nic (0 instances)	All	63992	Delete

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Home > Microsoft.LoadBalancer-20250710123413 | Overview >

frontend-lb

Load balancer

Search

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Overview

Activity log

Access control (IAM)

Tags

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Settings

Frontend IP configuration

Backend pools

Health probes

Load balancing rules

Inbound NAT rules

Outbound rules

Properties

Locks

Monitoring

Automation

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Essentials

JSON View

Resource group (move) : ITW-RG

Location : Central India

Subscription (move) : Azure for Students

Subscription ID : a93d05c3-3813-46d4-b354-abad66fa1940

SKU : Standard

Tags (edit) : Add tags

See more

Backend pool : 2 backend pools

Load balancing rule : http-rule (Tcp/77 to Tcp/80)

Health probe : http-probe (Http/80/ or /index.php)

Inbound NAT rules : None

Outbound rules : outbound-rule

Configure high availability and scalability for your applications

Create highly-available and scalable applications in minutes by using built-in load balancing for cloud services and virtual machines. Azure Load Balancer supports TCP/UDP-based protocols and protocols used for real-time voice and video messaging applications. [Learn more](#)

Balance IPv4 and IPv6 addresses

Native dual-stack endpoints help meet regulatory requirements and address the fast-growing number of devices in mobile and IoT. [Learn more](#)

View frontend IP configuration

Build highly reliable applications

Load Balancer improves application uptime by routing traffic to healthy nodes. [Learn more](#)

View health probes

Secure your networks

Control network traffic and protect private networks using built-in network address translation (NAT). [Learn more](#)

View inbound NAT rules

Step 4- Enabling Autoscaling on the frontend VMs to manage high traffic gracefully:

In this step, we implemented auto-scaling in Azure using a Virtual Machine Scale Set (VMSS) to ensure that our web application running on the frontend can scale automatically based on demand. First, we created a new VMSS within the same virtual network and subnet as our existing load balancer and frontend infrastructure. While creating the VMSS, we chose an existing image (custom or marketplace) to ensure all VM instances are identical in configuration. After successful deployment, we connected the VMSS to the backend pool of the load balancer, allowing it to receive traffic.

Next, we configured the Inbound NAT Pool on the load balancer to assign individual ports (like 50000, 50001, etc.) to each VMSS instance, enabling SSH access. For SSH to work, we ensured that port 22 was allowed in the inbound security rules of the load balancer and the Network Security Group (NSG) attached to the VMSS. After that, we manually SSH-ed into an instance using the public IP and NAT port, and installed Apache2 to test web server functionality.

Finally, we configured the auto-scaling rules based on CPU utilization. A rule was created to scale out (add a VM instance) when the CPU usage goes above a certain threshold (e.g., 70%) and scale in (remove an instance) when it goes below a defined level (e.g., 30%). This setup ensures our application is scalable, responsive under load, and cost-efficient during low usage periods.

Definitions:

- Virtual Machine Scale Set (VMSS): An Azure service that enables automatic deployment and management of a group of identical virtual machines for high availability and scalability.
- Auto-scaling: A process that automatically adjusts the number of VM instances in a scale set based on defined performance metrics or schedules.
- Load Balancer: A networking service that distributes incoming traffic across multiple VM instances to ensure even load and application reliability.
- Backend Pool: A group of VM or VMSS instances associated with a load balancer to receive and process network traffic.
- Inbound NAT Pool: A configuration that maps unique frontend ports on a load balancer to specific VMSS instances to allow individual access, such as SSH.
- Network Security Group (NSG): A set of firewall rules that controls inbound and outbound traffic to Azure resources, essential for enabling port access (e.g., SSH on port 22).
- Apache2: A widely used open-source HTTP server software installed to serve web pages from VM or VMSS instances.
- Custom Image: A user-defined VM image used to create uniform VMSS instances with preinstalled software and configurations.

azure portal - Search | frontend-lb - Microsoft Azure | frontend-vmss - Microsoft Azure | Apache2 Ubuntu Default Page: It's a Apache2 Ubuntu Default Page: It's a

https://portal.azure.com/#@aradhanamohanty247gmail.onmicrosoft.com/resource/subscriptions/a93d05c3-3813-46d4-b354-abad66fa840/resourceGroups/ITW-RG/provider... Search resources, services, and docs (G+)

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Home > Compute infrastructure | Virtual Machine Scale Set (VMSS) >

frontend-vmss Virtual machine scale set

Search

Move Start Restart Stop Hibernate Reimage Delete Refresh Feedback

Advisor (1 of 3): Enable automatic repair policy on Azure Virtual Machine Scale Sets (VMSS) →

Essentials

Resource group (move): ITW-RG
Status: 1 out of 1 succeeded
Location: Central India
Subscription (move): Azure for Students
Subscription ID: a93d05c3-3813-46d4-b354-abad66fa840

Operating system: Linux
Size: Standard_B1s (1 instance)
Public IP address: 4.247.143.151
Public IP address (IPv6): -
Virtual network/subnet: frontend-vnet/frontend-subnet
Orchestration mode: Uniform
Time created: 7/16/2025, 8:14 AM UTC

Tags (edit): Add tags

Properties Monitoring Capabilities (6) Recommendations (3) Tutorials

Virtual machine profile

Operating system: Linux
Capacity reservation group: -
Hibernation: Disabled

Azure Spot

Azure Spot: Disabled

Status

Provisioning state: Succeeded
Power state: 1 out of 1 running

Networking

Public IP address: 4.247.143.151
Public IP address (IPv6): -

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Microsoft Azure

Home > Compute infrastructure | Virtual Machine Scale Set (VMSS) > frontend-vmss

frontend-vmss | Load balancing Virtual machine scale set

Search

This is a new experience. Please provide feedback

+ Add load balancing Refresh Give feedback

Network interface / IP configuration

frontend-vnet-nic01 (primary) / frontend-vnet-nic01-defaultipConfiguration (primary)

Name	Type ↑	Frontend IP address	Frontend DNS address	Backend pool
frontend-lb	Load balancer	4.247.143.151	-	bepool

https://portal.azure.com/#@aradhanamohanty247gmail.onmicrosoft.com/resource/subscriptions/a93d05c3-3813-46d4-b354-abad66fa840/resourceGroups/ITW-RG/providers/Microsoft.Compute/virtualMachineScaleSets/frontend-vmss/LoadBalancing

azure portal - Search | frontend-lb - Microsoft Azure | frontend-vms - Microsoft Azure | Apache2 Ubuntu Default Page: It | x | +

https://portal.azure.com/#@aradhanamohanty247gmail.onmicrosoft.com/resource/subscriptions/a93d05c3-3813-46d4-b354-abad66faf840/resourceGroups/ITW-RG/provider...

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Home > frontend-lb

frontend-lb | Inbound NAT rules

Load balancer

Search | + Add | Refresh

Type to start filtering ...

Name	Frontend IP	Frontend port/range	Target	Service
frontend-lb-natRule01	4.247.143.151	50000 - 53599	bepool	SSH (TCP/22)

Give feedback

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azure portal - Search | frontend-lb - Microsoft Azure | frontend-vms - Microsoft Azure | basicNsgfrontend-vnet-nic01 - M | x | +

https://portal.azure.com/#@aradhanamohanty247gmail.onmicrosoft.com/resource/subscriptions/a93d05c3-3813-46d4-b354-abad66faf840/resourceGroups/ITW-RG/provider...

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Home > Network foundation | Network security groups

basicNsgfrontend-vnet-nic01

Network security group

Search | Move | Delete | Refresh | Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

- Inbound security rules
- Outbound security rules
- Network interfaces
- Subnets
- Properties
- Locks

Monitoring

- Alerts
- Diagnostic settings
- Logs

Add or remove favorites by pressing Ctrl+Shift+F

Essentials

Resource group (move): ITW-RG

Location: Central India

Subscription (move): Azure for Students

Subscription ID: a93d05c3-3813-46d4-b354-abad66faf840

Tags (edit): Add tags

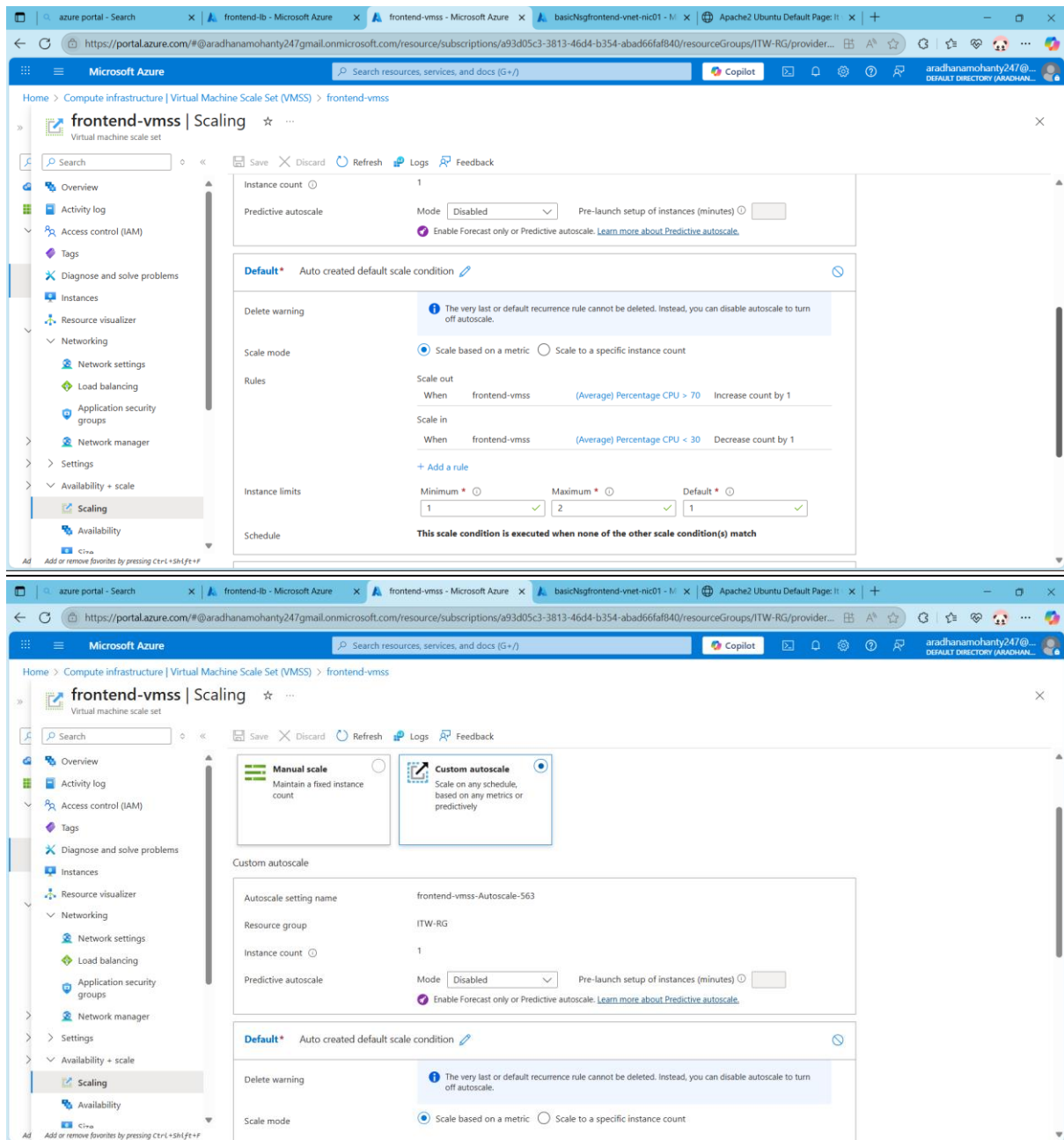
Custom security rules: 3 inbound, 1 outbound

Associated with: 0 subnets, 1 network interfaces

JSON View

Filter by name | Port == all | Protocol == all | Source == all | Destination == all | Action == all

Priority	Name	Port	Protocol	Source	Destination	Action
Inbound Security Rules						
100	AllowAnyCustom50000In-	50000-53599	Any	Any	Any	Allow
200	AllowAnyCustom22In-	22	Any	Any	Any	Allow
300	Tcp	80	Tcp	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerIn-	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny
Outbound Security Rules						
100	AllowAnyCustom443Out-	443	Any	Any	Any	Allow
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow



Step-5: Configure VPN access via the Jump VM, so all connections to the VMs happen through a private path.

Definitions:

- **Jump VM (Bastion Host):** A VM in a public subnet that acts as a gateway for accessing private VMs in a secure network.

- Private IP: IP address used within a virtual network. VMs in Azure Vnet can communicate via private IPs.
- SSH Proxy or Hop: The process of using the Jump VM to "hop" into other VMs using private IPs.
- VPN (Virtual Private Network): A secure tunnel allowing remote devices to connect to Azure network resources privately. (In your case, Jump VM is used instead of full VPN Gateway).

In Step 5, the objective was to securely access all virtual machines (frontend, backend, and VMSS instances) using private IP addresses only, by configuring a Jump VM (also known as a bastion host). This Jump VM was placed within the same virtual network (Vnet) as the other VMs and was assigned a public IP address with port 22 open for SSH access. Once connected to the Jump VM via SSH from the local system, we were able to access other internal VMs through their private IPs, ensuring that these VMs do not need public IPs or exposed ports. This step simulates VPN-like secure internal connectivity, where the Jump VM acts as a controlled gateway for managing resources within the private network, reducing attack surface and improving security posture for the cloud deployment.

The screenshot displays the Microsoft Azure portal interface for a virtual machine named 'jump-server'. The left sidebar shows the navigation menu with options like Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Connect, Bastion, Networking, Load balancing, Application security groups, Network manager, Settings, and Disks. The main content area shows the 'Overview' tab for the 'jump-server' VM. Key details include:

- Resource group:** ITW-RG
- Status:** Running
- Location:** Central India
- Subscription:** Azure for Students
- Subscription ID:** a93d05c3-3813-46d4-b354-abad66faf840
- Operating system:** Linux (ubuntu 24.04)
- Size:** Standard B1s (1 vcpu, 1 GiB memory)
- Public IP address:** 4.247.135.206
- Virtual network/subnet:** frontend-vnet/jump-subnet
- DNS name:** Not configured
- Health state:** -
- Time created:** 7/16/2025, 7:32 AM UTC

Below the overview, there are sections for 'Properties', 'Monitoring', 'Capabilities (7)', 'Recommendations (4)', and 'Tutorials'. The 'Virtual machine' section shows the computer name 'jump-server', operating system 'Linux (ubuntu 24.04)', and VM generation 'V2'. The 'Networking' section shows the public IP address '4.247.135.206' (Network interface jump-server605), public IP address (IPv6) '-', and private IP address '10.0.2.4'.

Microsoft Azure portal showing the Network settings for two virtual machines: **jump-server** and **frontend-vm**.

jump-server | Network settings

Network interface / IP configuration: **jump-server605 (primary) / ipconfig1 (primary)**

Essentials

Network interface	: jump-server605	Load balancers	: 0 (Configure)
Virtual network / subnet	: frontend-vnet / jump-subnet	Application security gro...	: 0 (Configure)
Public IP address	: 4.247.135.206	Network security group	: jump-server-nsg
Private IP address	: 10.0.2.4	Accelerated networking	: Disabled
Admin security rules	: 0 (Configure)	Effective security rules	: 0

Rules Collapse all

Network security group **jump-server-nsg** (attached to networkInterface: jump-server605)
Impacts 0 subnets, 1 network interfaces

frontend-vm | Network settings

Network interface / IP configuration: **frontend-vm400 (primary) / ipconfig1 (primary)**

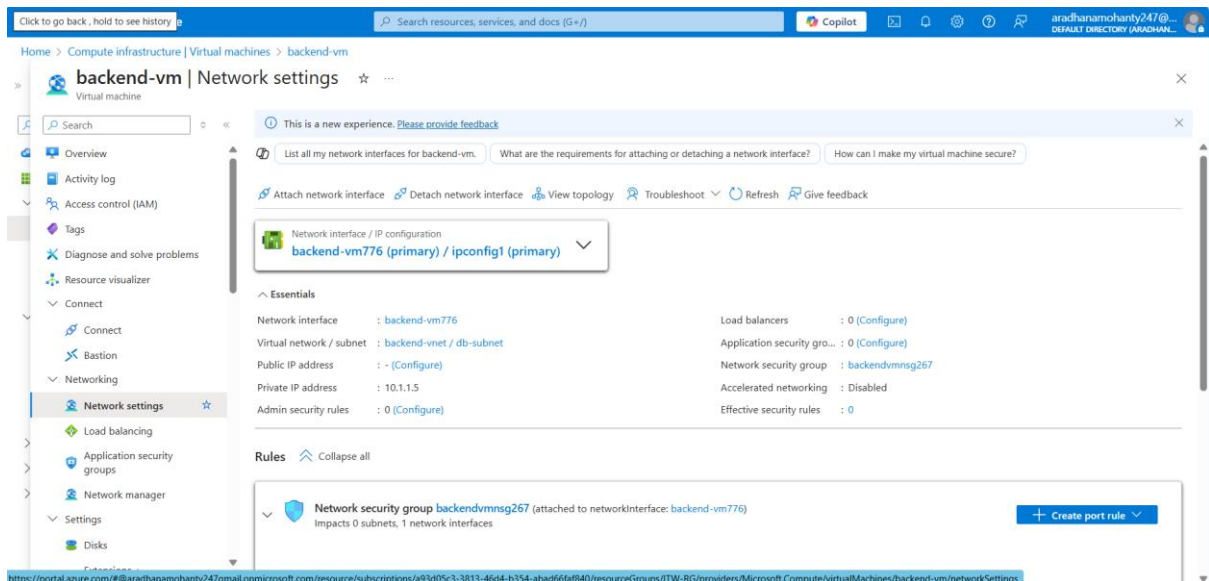
Essentials

Network interface	: frontend-vm400	Load balancers	: 0 (Configure)
Virtual network / subnet	: frontend-vnet / frontend-subnet	Application security gro...	: 0 (Configure)
Public IP address	: 4.247.138.80	Network security group	: frontend-vm-nsg
Private IP address	: 10.0.1.4	Accelerated networking	: Disabled
Admin security rules	: 0 (Configure)	Effective security rules	: 0

Rules Collapse all

Network security group **frontend-vm-nsg** (attached to networkInterface: frontend-vm400)
Impacts 0 subnets, 1 network interfaces

S



```
mca@ARADHANA01_07:~$ ssh siliconuser123@4.247.135.206
siliconuser123@4.247.135.206's password:
Permission denied, please try again.
siliconuser123@4.247.135.206's password:
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.11.0-1018-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Jul 16 14:50:19 UTC 2025

System load:  0.0               Processes:            115
Usage of /:   5.6% of 28.02GB   Users logged in:     0
Memory usage: 30%              IPv4 address for eth0: 10.0.2.4
Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Wed Jul 16 10:50:57 2025 from 157.49.157.15
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

siliconuser123@jump-server:~$
```

```
siliconuser123@jump-server:~$ ssh siliconuser123@10.0.1.5
The authenticity of host '10.0.1.5 (10.0.1.5)' can't be established.
ED25519 key fingerprint is SHA256:sxZJLVopn2IkJYMBM92VrwD4g0jffv1J1Gux1SsMaYo.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.1.5' (ED25519) to the list of known hosts.
siliconuser123@10.0.1.5's password:
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.11.0-1018-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Jul 16 15:20:01 UTC 2025

System load:  0.0                Processes:    114
Usage of /:   6.5% of 28.02GB    Users logged in: 1
Memory usage: 28%               IPv4 address for eth0: 10.0.1.5
Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.
10 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Jul 16 13:47:32 2025 from 152.58.103.220
siliconuser123@frontend-000000:~$ exit
siliconuser123@frontend-000000:~$ exit
logout
Connection to 10.0.1.5 closed.
siliconuser123@jump-server:~$ ssh siliconuser123@10.1.1.5
siliconuser123@10.1.1.5's password:
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.11.0-1018-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Jul 16 15:21:00 UTC 2025

System load:  0.0                Processes:    112
Usage of /:   5.6% of 28.02GB    Users logged in: 0
Memory usage: 30%               IPv4 address for eth0: 10.1.1.5
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Wed Jul 16 07:40:12 2025 from 10.0.2.4
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

siliconuser123@backend-vm:~$ █
```

Step -6: Use Azure Hosting service to deploy your application and map your domain name to the Public IP of the load balancer.

Definition:

- Custom Domain Mapping: It is the process of linking a domain name purchased from a provider like GoDaddy to a cloud-based public IP (such as an Azure Load Balancer IP), allowing users to access the deployed application using a user-friendly domain (e.g., www.itwsit.xyz) instead of a numeric IP address.
- DNS (Domain Name System): A hierarchical naming system that translates human-readable domain names (like www.example.com) into IP addresses (like 20.198.10.15) that computers use to identify each other on the network.
- A Record (Address Record): A DNS record that maps a domain name to an IPv4 address. In Step 6, we used it to point the domain to the frontend Load Balancer's public IP.
- GoDaddy DNS Management: A dashboard provided by GoDaddy (domain registrar) where you manage DNS records such as A, CNAME, and MX to control how your domain connects to online resources.
- Propagation Time: The amount of time it takes for DNS changes (like adding or updating an A record) to be updated and recognized across the global DNS network. It can take a few minutes to several hours.
- Apache Web Server: A widely used open-source web server that serves HTTP content. It was installed on the VMSS instance to deliver a test web page linked to the domain.

In Step 6, we integrated a custom domain purchased from GoDaddy with our deployed web application on Azure. We pointed the domain to the frontend Load Balancer's public IP using an A record in the DNS settings. After that, we tested DNS resolution and ensured that Apache was successfully serving the webpage from one of the VMSS instances. We also customized the web page using a simple HTML update to confirm the end-to-end connection between domain and cloud infrastructure.

https://dcc.godaddy.com/control/dnsmanagement?domainName=itwsit.xyz&subtab=dns-records

Domains

Portfolio

DNS

Transfers

Services

Settings

Activity Log

DNS Management

☆ Your domain is in demand

Other GoDaddy customers are searching for **itwsit.xyz**! Add Domain Protection to ensure you don't lose this pop

[Add Protection](#) →

itwsit.xyz

Domain Settings

Use My Domain

DNS Records

Forwarding

Nameservers

Premium DNS

Hostnames

DNSSEC

Add a new record

DNS records define how your domain behaves, like showing your website content and delivering your email.

Easily verify domain ownership

Need to verify ownership of your domain to connect to an external service? We've made it easier than ever.

Create MX Records

Quickly create MX records to connect your domain with

AM

Aradhana Mohanty

Customer #: 664414808

PIN: View

My Products

Account Settings

Renewals & Billing

Shop GoDaddy.com

Sign Out

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https://dcc.godaddy.com/control/dnsmanagement?domainName=itwsit.xyz&subtab=dns-records

Domains

Portfolio

DNS

Transfers

Services

Settings

Activity Log

DNS Records

Forwarding

Nameservers

Premium DNS

Hostnames

DNSSEC

Crypto Wallet

Add a new record

DNS records define how your domain behaves, like showing your website content and delivering your email.

Add New Record

Easily verify domain ownership

Need to verify ownership of your domain to connect to an external service? We've made it easier than ever.

Verify Domain Ownership

Create MX Records

Quickly create MX records to connect your domain with email services.

Connect Now

Filters

Actions

Type	Name	Data	TTL	Delete	Edit
A	@	4.247.143.151	600 seconds		

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

aradhana.mohanty247@... DEFAULT DIRECTORY

Home > frontend-lb

frontend-lb | Frontend IP configuration

Load balancer

Search

+ Add Refresh

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Frontend IP configuration

Backend pools

Health probes

Load balancing rules

Inbound NAT rules

Outbound rules

Properties

Locks

Monitoring

The frontend IP address configuration of a load balancer serves as the entry point for incoming traffic to the load balancer, and the load balancer then distributes the traffic to the backend pool of virtual machines or services. [Learn more](#)

Type to start filtering ...

Showing all 1 items

Name	IP address	Rules count
frontend-lb-frontendconfig01	4.247.143.151 (frontend-lb-publicip)	3

Give feedback

https://portal.azure.com/#@aradhana.mohanty247@mail.onmicrosoft.com/resource/subscriptions/a93d05c3-3813-4604-b354-abad66f8f840/resourceGroups/ITW-RG/providers/Microsoft.Network/loadBalancers/frontend-lb/frontendipPool

```
siliconuser123@frontend-000000: /var/www/html
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split Multitex Tunneling Packages Settings Help
Quick co 2 siliconuser123@frontend-000000
mc@99ADAHAMARI 07:~$ ssh siliconuser123@4.247.143.151 -p 50000
siliconuser123@4.247.143.151:~$
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.11.0-1010-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Jul 18 06:36:34 UTC 2025

System load: 0.0          Processes: 114
Usage of /: 6.5% of 28.02GB Users logged in: 1
Memory usage: 36%        IPv4 address for eth0: 10.0.1.5
Swap usage: 0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.
   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

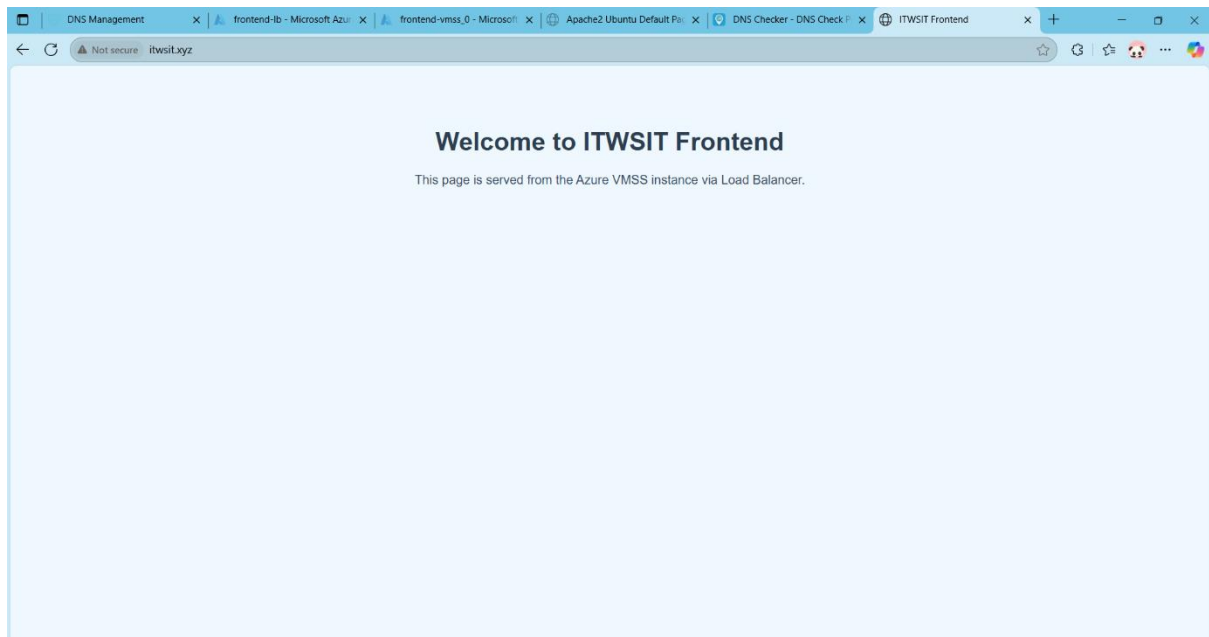
13 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

*** System restart required ***
Last login: Wed Jul 16 15:20:01 2025 from 10.0.2.4
siliconuser123@frontend-000000:~$ sudo apt update
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
17 packages can be upgraded. Run 'apt list --upgradable' to see them.
siliconuser123@frontend-000000:~$ sudo apt install apache2 -y
Reading package lists... Done

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siliconuser123@frontend-000000: /var/www/html
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split Multitex Tunneling Packages Settings Help
Quick co 2 siliconuser123@frontend-000000
Reading state information... Done
apache2 is already the newest version (2.4.58-1ubuntu0.7).
0 upgraded, 0 newly installed, 0 to remove and 17 not upgraded.
siliconuser123@frontend-000000:~$ sudo systemctl start apache2
siliconuser123@frontend-000000:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
siliconuser123@frontend-000000:~$ cd /var/www/html
siliconuser123@frontend-000000:/var/www/html$ sudo rm index.html
siliconuser123@frontend-000000:/var/www/html$ scp -P 50000 path/to/your/index.html azureuser@4.247.143.151:/var/www/html/
scp: stat local 'path/to/your/index.html': No such file or directory
siliconuser123@frontend-000000:/var/www/html$ scp -p 50000 path/to/your/index.html azureuser@4.247.143.151:/var/www/html/
scp: stat local '50000': No such file or directory
siliconuser123@frontend-000000:/var/www/html$ echo "<h1 style='color: green;'>Welcome to ITWSIT Frontend via VMSS</h1>" | sudo tee index.html
<h1 style='color: green;'>Welcome to ITWSIT Frontend via VMSS</h1>
siliconuser123@frontend-000000:/var/www/html$ sudo tee /var/www/html/index.html > /dev/null <<EOF
<!DOCTYPE html>
<html>
<head>
  <title>ITWSIT Frontend</title>
  <style>
    body {
      background-color: #f0f8ff;
      font-family: Arial, sans-serif;
      text-align: center;
      padding-top: 50px;
    }
    h1 {
      color: #2c3e50;
    }
    p {
      color: #34495e;
    }
  </style>
</head>
<body>
  <h1>Welcome to ITWSIT Frontend</h1>
  <p>This page is served from the Azure VMSS instance via Load Balancer.</p>
</body>
</html>
EOF
siliconuser123@frontend-000000:/var/www/html$
```



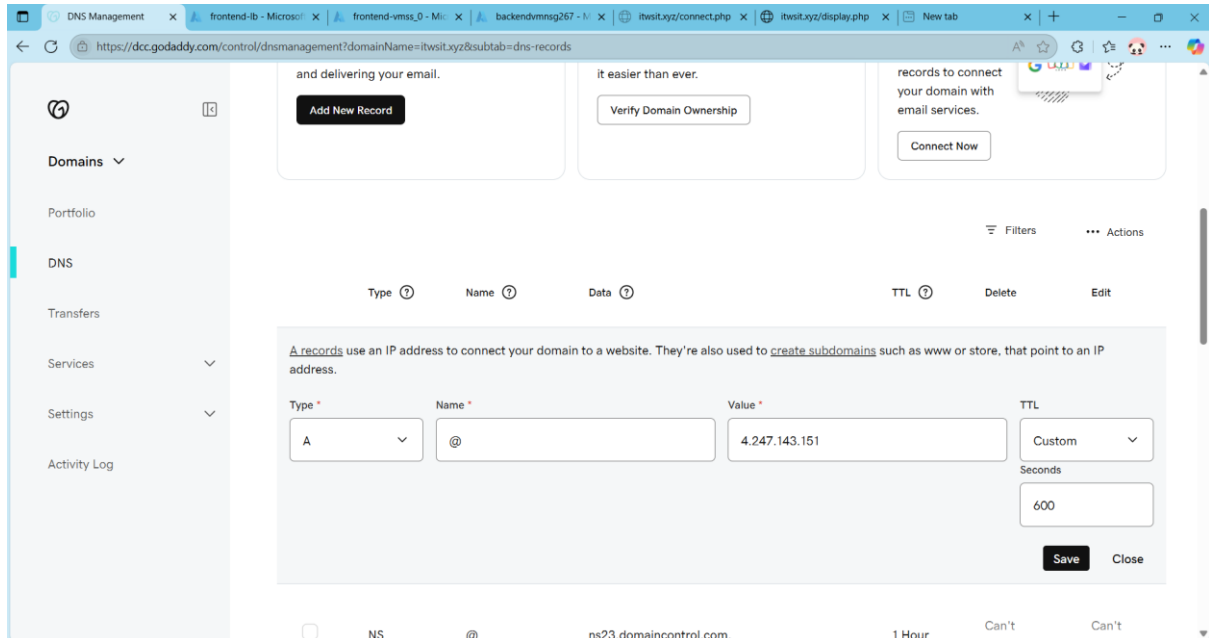
Step -7: Perform the sample user operations from the frontend, and verify that changes reflected in the backend database.

Definitions:

- DNS: A Record Maps a domain name to an IP address.
- Apache: Web server software that delivers web content.
- PHP: Server-side scripting language for dynamic content.
- MySQL: Open-source relational database to store form data.
- Load Balancer: Routes traffic to VMSS instances.
- VMSS: Auto-scaling group of identical frontend web VMs.
- Backend VM: Private VM hosting the database.
- Form Submission: HTML + PHP code to collect and save user input.

In Step 7, I linked the frontend application with a custom domain using Azure DNS and Load Balancer public IP. I installed and configured Apache on VMSS instances, set up PHP support, and verified connectivity with test pages. I then created a complete data flow system: a form that takes input, stores it in a MySQL database hosted on the backend VM, and displays the records. This step finalizes the full-stack connectivity between frontend, backend, and database, accessible through the domain.

1. DNS Integration: Mapped your domain **itwsit.xyz** to the Load Balancer public IP. Configured GoDaddy DNS A record correctly
2. Apache & PHP Configuration: Installed and verified Apache + PHP in the VMSS instance.



```
siliconuser123@frontend-00000000:~$ sudo apt install apache2 php libapache2-mod-php php
p-mysql -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.58-1ubuntu8.7).
php is already the newest version (2:8.3+93ubuntu2).
The following additional packages will be installed:
  php8.3-mysql
The following NEW packages will be installed:
  libapache2-mod-php php-mysql php8.3-mysql
0 upgraded, 3 newly installed, 0 to remove and 17 not upgraded.
Need to get 133 kB of archives.
After this operation, 488 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libapache2-mod-php all
2:8.3+93ubuntu2 [4224 B]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-mysql a
md64 8.3.6-0ubuntu0.24.04.5 [127 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 php-mysql all 2:8.3+93
ubuntu2 [1838 B]
Fetched 133 kB in 0s (540 kB/s)
Selecting previously unselected package libapache2-mod-php.
(Reading database ... 68765 files and directories currently installed.)
Preparing to unpack .../libapache2-mod-php_2%3a8.3+93ubuntu2_all.deb ...
Unpacking libapache2-mod-php (2:8.3+93ubuntu2) ...
Selecting previously unselected package php8.3-mysql.
Preparing to unpack .../php8.3-mysql_8.3.6-0ubuntu0.24.04.5_amd64.deb ...
Unpacking php8.3-mysql (8.3.6-0ubuntu0.24.04.5) ...
Selecting previously unselected package php-mysql.
Preparing to unpack .../php-mysql_2%3a8.3+93ubuntu2_all.deb ...
Unpacking php-mysql (2:8.3+93ubuntu2) ...
Setting up php8.3-mysql (8.3.6-0ubuntu0.24.04.5) ...

Creating config file /etc/php/8.3/mods-available/mysqlnd.ini with new version
Creating config file /etc/php/8.3/mods-available/mysqli.ini with new version
Creating config file /etc/php/8.3/mods-available/pdo_mysql.ini with new version
Setting up libapache2-mod-php (2:8.3+93ubuntu2) ...
Setting up php-mysql (2:8.3+93ubuntu2) ...

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```

2. MySQL Setup

- Installed MySQL on the backend VM.
- Created database **itwsit_db** and table users.


```
siliconuser123@backend-vm:~$ sudo mysql
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.42-0ubuntu0.24.04.2 (Ubuntu)

Copyright (c) 2000, 2025, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE itwst;
Query OK, 1 row affected (0.04 sec)

mysql>
mysql> USE itwst;
Database changed
mysql> CREATE TABLE users (
  ->   id INT AUTO_INCREMENT PRIMARY KEY,
  ->   name VARCHAR(100),
  ->   email VARCHAR(100),
  ->   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
  -> );
Query OK, 0 rows affected (0.08 sec)

mysql> INSERT INTO users (name, email) VALUES ('Test User', 'test@example.com');
Query OK, 1 row affected (0.02 sec)

mysql> SELECT * FROM users;
+----+-----+-----+-----+
| id | name  | email          | created_at |
+----+-----+-----+-----+
| 1  | Test User | test@example.com | 2025-07-10 00:51:43 |
+----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> CREATE USER 'webuser'@'%' IDENTIFIED BY 'webuser0123';
Query OK, 0 rows affected (0.02 sec)

mysql> GRANT ALL PRIVILEGES ON itwst.* TO 'webuser'@'%';

--- itwst.xyz ping statistics ---
1100 packets transmitted, 0 received, 100% packet loss, time 1125389ms

siliconuser123@frontend-000000:~$ sudo nano /var/www/html/connect.php
siliconuser123@frontend-000000:~$ sudo systemctl restart apache2
siliconuser123@frontend-000000:~$ ping 10.1.1.5
PING 10.1.1.5 (10.1.1.5) 56(84) bytes of data:
64 bytes from 10.1.1.5: icmp_seq=1 ttl=64 time=1.60 ms
64 bytes from 10.1.1.5: icmp_seq=2 ttl=64 time=0.967 ms
64 bytes from 10.1.1.5: icmp_seq=3 ttl=64 time=1.15 ms
64 bytes from 10.1.1.5: icmp_seq=4 ttl=64 time=0.980 ms
64 bytes from 10.1.1.5: icmp_seq=5 ttl=64 time=1.08 ms
64 bytes from 10.1.1.5: icmp_seq=6 ttl=64 time=0.977 ms
64 bytes from 10.1.1.5: icmp_seq=7 ttl=64 time=1.11 ms
64 bytes from 10.1.1.5: icmp_seq=8 ttl=64 time=1.01 ms
64 bytes from 10.1.1.5: icmp_seq=9 ttl=64 time=1.05 ms
64 bytes from 10.1.1.5: icmp_seq=10 ttl=64 time=1.05 ms
64 bytes from 10.1.1.5: icmp_seq=11 ttl=64 time=1.15 ms
64 bytes from 10.1.1.5: icmp_seq=12 ttl=64 time=1.15 ms
64 bytes from 10.1.1.5: icmp_seq=13 ttl=64 time=1.08 ms
64 bytes from 10.1.1.5: icmp_seq=14 ttl=64 time=1.37 ms
64 bytes from 10.1.1.5: icmp_seq=14 ttl=64 time=1.44 ms
^C
--- 10.1.1.5 ping statistics ---
14 packets transmitted, 14 received, 0% packet loss, time 13017ms
rtt min/avg/max/mdev = 0.967/1.158/1.600/0.184 ms
siliconuser123@frontend-000000:~$ sudo netstat -tulnp |grep 3306
sudo: netstat: command not found
siliconuser123@frontend-000000:~$ sudo nano /var/www/html/connect.php
siliconuser123@frontend-000000:~$ curl https://localhost/test.php
curl: (7) Failed to connect to localhost port 443 after 0 ms: Couldn't connect to se
rver
siliconuser123@frontend-000000:~$ curl https://localhost/connect.php
curl: (7) Failed to connect to localhost port 443 after 0 ms: Couldn't connect to se
rver
siliconuser123@frontend-000000:~$ curl http://localhost/connect.php
<h2>Starting Database Connection Test...</h2>Connecting to 10.1.1.5 ...<br><span sty
le= color:green;>☑ Connected successfully to the database.</span>siliconuser123@fr
ontend-000000:~$ sudo nano /etc/hosts
sudo nano /var/www/html/index.php
siliconuser123@frontend-000000:~$ sudo nano /var/www/html/index.html
siliconuser123@frontend-000000:~$ sudo nano /var/www/html/submit.php
siliconuser123@frontend-000000:~$ sudo nano /var/www/html/display.php
siliconuser123@frontend-000000:~$
```

4. Form Creation + PHP Scripts

- Created:
- index.html: A user form (name, email).
- connect.php: Handles DB connection.
- submit.php: Saves form data to DB.
- display.php: Shows saved entries from DB.

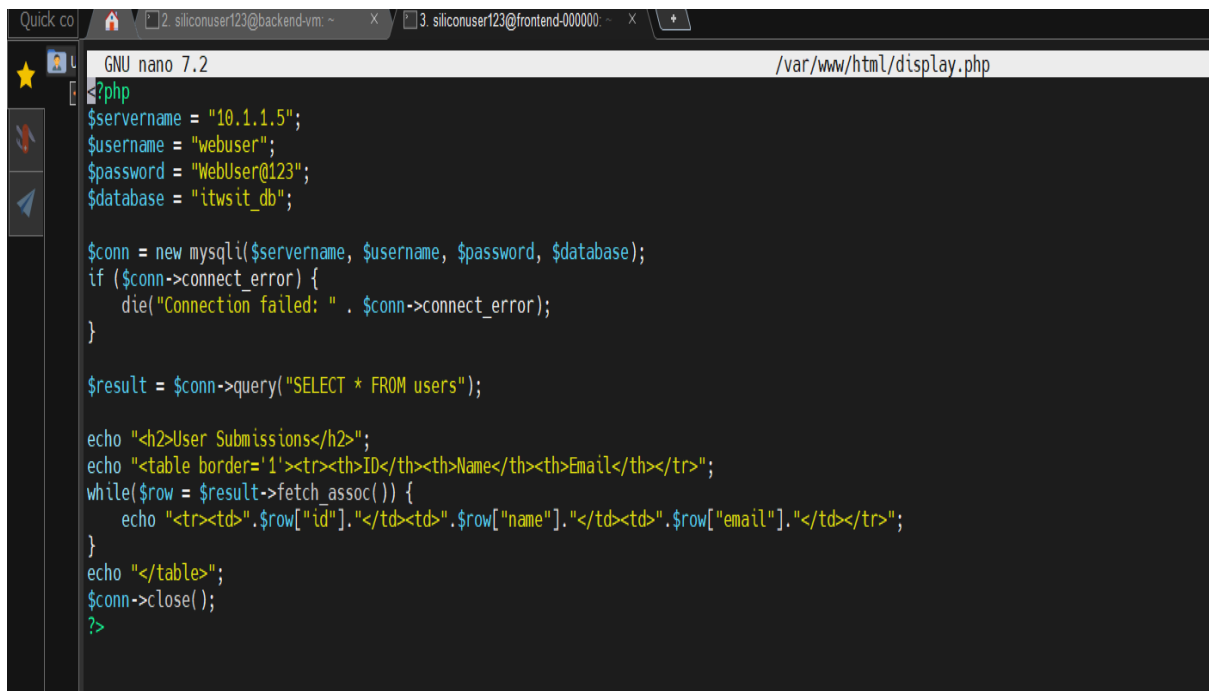
```
Quick co 2. siliconuser123@backend-vm: ~ 3. siliconuser123@frontend-000000: ~ +
GNU nano 7.2
<!DOCTYPE html>
<html>
<head>
  <title>ITWSIT Form</title>
</head>
<body>
  <h1>Submit Your Info</h1>
  <form method="POST" action="submit.php">
    Name: <input type="text" name="name" required><br><br>
    Email: <input type="email" name="email" required><br><br>
    <input type="submit" value="Submit">
  </form>
  <br><a href="display.php">View Submissions</a>
</body>
</html>

GNU nano 7.2 /va
<?php
$servername = "10.1.1.5"; // Backend VM private IP
$username = "webuser";
$password = "WebUser@123";
$dbname = "itwsit_db";

$conn = new mysqli($servername, $username, $password, $dbname);
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

$name = $_POST['name'];
$email = $_POST['email'];

$sql = "INSERT INTO users (name, email) VALUES ('$name', '$email')";
if ($conn->query($sql) === TRUE) {
    echo "New record added successfully. <a href='index.html'>Go Back</a>";
} else {
    echo "Error: " . $conn->error;
}
$conn->close();
?>
```



```
GNU nano 7.2 /var/www/html/display.php
<?php
$servername = "10.1.1.5";
$username = "webuser";
$password = "WebUser@123";
$dbname = "itwsit_db";

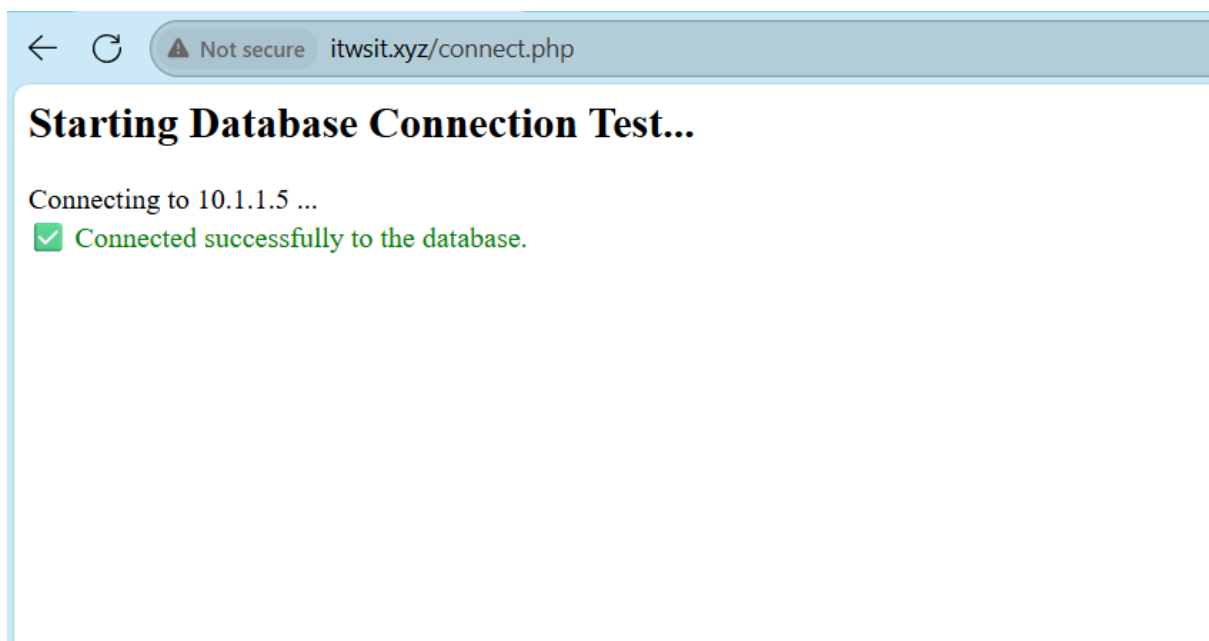
$conn = new mysqli($servername, $username, $password, $dbname);
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

$result = $conn->query("SELECT * FROM users");

echo "<h2>User Submissions</h2>";
echo "<table border='1'><tr><th>ID</th><th>Name</th><th>Email</th></tr>";
while($row = $result->fetch_assoc()) {
    echo "<tr><td>".$row["id"]."</td><td>".$row["name"]."</td><td>".$row["email"]."</td></tr>";
}
echo "</table>";
$conn->close();
?>
```

5. Verification

- Connected successfully to the database via connect.php
- Form submits data successfully
- Data is stored and displayed from backend VM via display.php.



ITWSIT Form

Not secure itwsit.xyz/index.html

Submit Your Info

Name:

Email:

[View Submissions](#)

itwsit.xyz/submit.php

Not secure itwsit.xyz/submit.php

New record added successfully. [Go Back](#)

itwsit.xyz/display.php

Not secure itwsit.xyz/display.php

User Submissions

ID	Name	Email
1	Test User	test@example.com
2	Aradhana Mohanty	aradhanamohanty247@gmail.com
3	Dimpy Mohanty	dimpymohanty2010@gmail.com

Step:8- Apply an SSL certificate to your domain to ensure secure access over HTTPS.

Definitions:

- Certbot: A tool to get free SSL certificates from Let's Encrypt.
- SSL/TLS: Encrypts the data between browser and server (used by HTTPS).
- Apache Virtual Host: Configuration block that defines how Apache responds to a domain or port.
- Port 443: Default port used by HTTPS for secure traffic.
- NSG: (Network Security Group) Controls inbound/outbound traffic to Azure resources.
- Load Balancer Rule: Maps public traffic (like port 443) to backend VM/VMSS.

In Step 7 of our Azure project, we secured our custom domain itwsit.xyz by installing a Let's Encrypt SSL certificate on the frontend VMSS using Certbot with Apache. This allows our web application to be served over HTTPS, ensuring secure communication between clients and the server. We first verified that Apache was properly configured with a virtual host for port 443. Then we executed `sudo certbot --apache` and followed the prompts to issue an SSL certificate. After successful installation, we restarted Apache and confirmed the presence of the certificate files. We verified HTTPS access through the browser. Finally, since auto-renewal was optional in the project's scope, we disabled any automated renewal mechanisms (like systemd timers or cron jobs) and opted for manual renewal closer to the expiry date (October 16, 2025). This step ensures that users visiting our web app experience a secure, trusted connection.

1. Installed Certbot on Apache (on the VMSS instance)

- `sudo apt install certbot python3-certbot-apache.`
- Generated a Free Let's Encrypt SSL Certificate.
- `sudo certbot --Apache`

```

siliconuser123@frontend-000000:~$ curl -I https://itwsit.xyz
HTTP/1.1 200 OK
Date: Sun, 20 Jul 2025 06:57:16 GMT
Server: Apache/2.4.58 (Ubuntu)
Last-Modified: Fri, 18 Jul 2025 11:17:02 GMT
ETag: "17e-63a3243e677e7"
Accept-Ranges: bytes
Content-Length: 382
Vary: Accept-Encoding
Content-Type: text/html

siliconuser123@frontend-000000:~$ sudo apt install certbot python3-certbot-apache
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
certbot is already the newest version (2.9.0-1).
python3-certbot-apache is already the newest version (2.9.0-1).
0 upgraded, 0 newly installed, 0 to remove and 17 not upgraded.
siliconuser123@frontend-000000:~$ sudo certbot --apache
Saving debug log to /var/log/letsencrypt/letsencrypt.log

Which names would you like to activate HTTPS for?
We recommend selecting either all domains, or all domains in a VirtualHost/server block.
.....
1: itwsit.xyz
.....

```

```

siliconuser123@frontend-000000: ~
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help

Quick co [2. siliconuser123@frontend-000000: ~] x
System information as of Sun Jul 20 06:50:20 UTC 2025

System load: 0.0          Processes: 126
Usage of /: 6.9% of 28.02GB Users logged in: 1
Memory usage: 37%        IPv4 address for eth0: 10.0.1.5
Swap usage: 0%

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
just raised the bar for easy, resilient and secure K8s cluster deployment.
https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

17 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

*** System restart required ***
Last login: Fri Jul 18 12:23:16 2025 from 157.49.154.194
siliconuser123@frontend-000000:~$ sudo apache2ctl -S
VirtualHost configuration:
*:443          itwsit.xyz (/etc/apache2/sites-enabled/000-default-le-ssl.conf:2)
*:80          frontend-000000.internal.cloudapp.net (/etc/apache2/sites-enabled/000-default.conf:1)
ServerRoot: "/etc/apache2"
Main DocumentRoot: "/var/www/html"
Main ErrorLog: "/var/log/apache2/error.log"
Mutex ssl-cache: using_defaults
Mutex default: dir="/var/run/apache2/" mechanism=default
Mutex mpm-accept: using_defaults
Mutex watchdog-callback: using_defaults
Mutex rewrite-map: using_defaults
Mutex ssl-stapling-refresh: using_defaults
Mutex ssl-stapling: using_defaults
Pidfile: "/var/run/apache2/apache2.pid"
Define: DUMP_VHOSTS
Define: DUMP_RUN_CFG
User: name="www-data" id=33
Group: name="www-data" id=33

```

2.Azure NSG inbound rule for port 443 was added (priority 180,).

3.Azure Load Balancer rule for port 443 was checked/added to allow HTTPS traffic to VMSS.

basicNsgfrontend-vnet-nic01 | Inbound security rules

Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority. [Learn more](#)

Priority	Name	Port	Protocol	Source	Destination	Action
100	AllowAnyCustom50000In...	50000-53599	Any	Any	Any	Allow
180	AllowAnyCustom443Inbo...	443	TCP	Any	Any	Allow
200	AllowAnyCustom22Inbo...	22	Any	Any	Any	Allow
300	Tcp	80	Tcp	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancer...	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny

frontend-lb | Load balancing rules

A load balancer rule is used to define how incoming traffic is distributed to all the instances within the backend pool. A load-balancing rule maps a given frontend IP configuration and port to multiple backend IP addresses and ports. An example would be a rule created on port 80 to load balance web traffic. [Learn more](#)

Name	Protocol	Backend pool	Health probe	Health status
frontend-lb-lbrule01	TCP/80	bepool	frontend-lb-probe01	View details
LBRule-HTTPS	TCP/443	bepool	health-frontend	View details

4. Tested HTTPS access using:

Browser (<https://itwsit.xyz>)

Terminal (`curl -I https://itwsit.xyz`)

ITWSIT Form

https://itwsit.xyz

Submit Your Info

Name:

Email:

[View Submissions](#)

itwsit.xyz/submit.php

https://itwsit.xyz/submit.php

New record added successfully. [Go Back](#)

itwsit.xyz/display.php

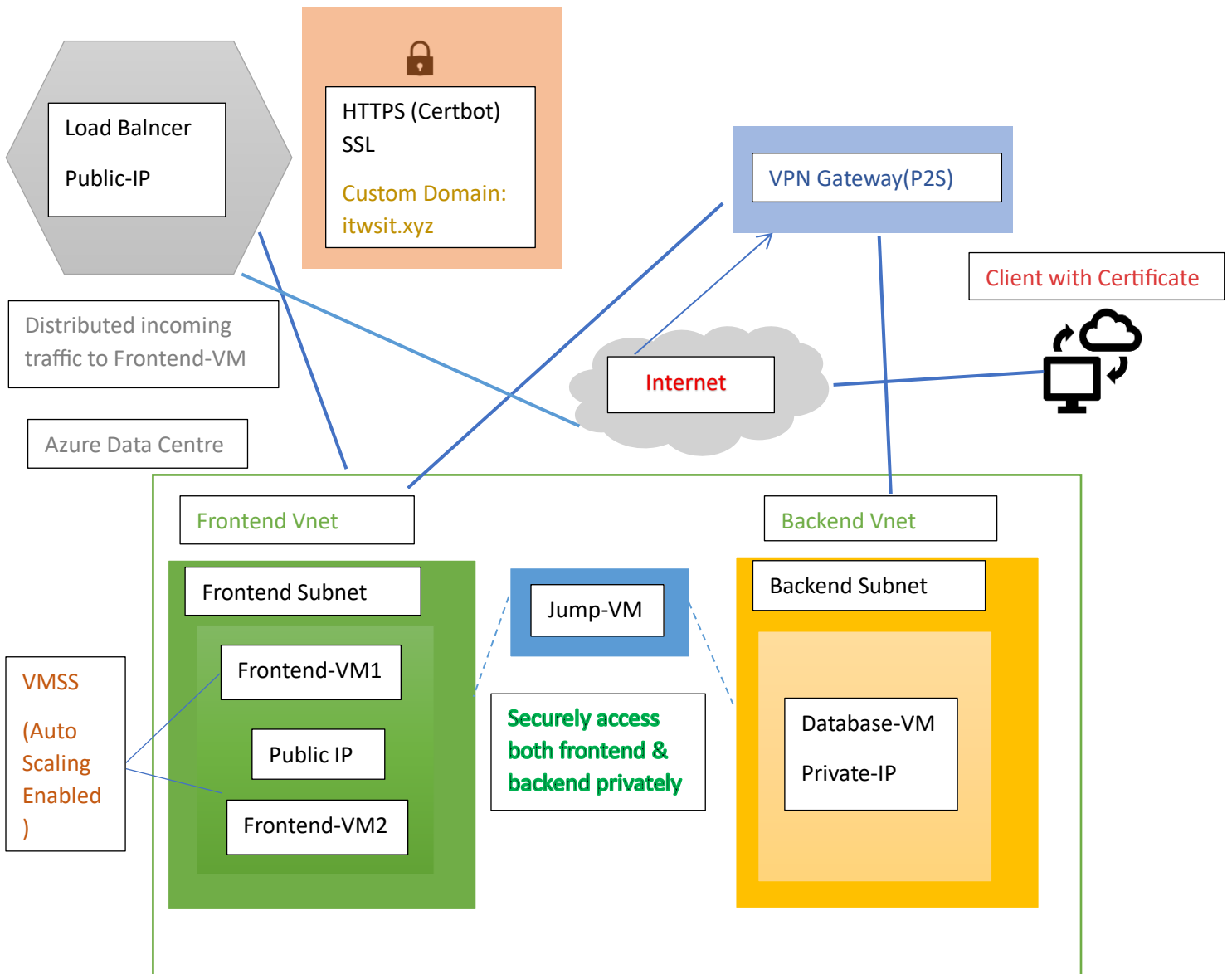
basicNsgfrontend-vnet-nic01 - M

https://itwsit.xyz/display.php

User Submissions

ID	Name	Email
1	Test User	test@example.com
2	Aradhana Mohanty	aradhanamohanty247@gmail.com
3	Dimpy Mohanty	dimpymohanty2010@gmail.com
4	Tanu	tanu123@gmail.com
5	silicon	silicon123@gmail.com

Architecture of the Project



Title: Full Stack App Development with VNet, VM, Load Balancer & Domain-Azure Cloud Project.

In this project, I deployed a full-stack web application on Microsoft Azure following a production-like architecture. I started by creating two separate Virtual Networks (VNETs) — one for the frontend and one for the backend/database. This separation helps improve security and manageability by isolating traffic between tiers.

Within each VNet, I deployed Virtual Machines. For the frontend, I created a scalable VMSS (Virtual Machine Scale Set) so that it can handle more users when needed. For the backend, I deployed a standard VM hosting the database. I also added a Jump VM in a secure zone to act as a gateway for internal SSH connections, avoiding exposing VMs directly to the internet.

To handle incoming traffic, I placed a Load Balancer in front of the frontend VMSS. Only the Load Balancer has a public IP, while the actual VMs remain private. I then enabled autoscaling on the frontend VMSS to ensure it automatically adds or removes instances based on CPU load. This approach keeps the app responsive and cost-effective.

To securely connect to the Azure network, I configured a VPN Gateway and used Point-to-Site (P2S) VPN. I generated client and root certificates to ensure that only trusted clients can access internal resources. This setup follows industry best practices by preventing open SSH/RDP access.

After that, I mapped a custom domain (itwsit.xyz) to the public IP of the Load Balancer, and finally applied an SSL certificate using Certbot to enable HTTPS. This ensures encrypted traffic and better trust for users accessing the application. I chose not to auto-renew the certificate for now, since it's a demo deployment.

Overall, I followed cloud best practices like resource separation, minimal public exposure, VPN access for security, scalability, and SSL for secure communication, making this a reliable and realistic production-ready Azure deployment.

THANK YOU!